

5M P&D Zone Rendering Fix — Report

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File Modified: smc_premium_discount_indicator_v17.54.pine

Lines Changed: 2035-2137 (zone section only)

Railway:  Deployed & Healthy

Problem

On 5M charts, only **one Supply zone** was visible and **no Demand zones** were painting. H4 Premium/Discount background zones worked correctly (separate system).

Root Cause Analysis

A detailed diff of **all 11 Pine Script versions** (v17.15 through v17.54) revealed that the Supply/Demand zone code was **identical** across every version. The zone rendering used **scalar var variables**:

```
// OLD (broken)  scalar tracking, ONE zone per type
var float demandTop    = na      //  overwritten each detection
var float demandBot    = na
var int  demandBar     = 0
var bool demandActive  = false
var box  demandBox     = na      //  only ONE box ever drawn
```

Each time a new swing point was detected, it **overwrote** the previous zone data. Only the most recent zone of each type could ever render. On 5M charts (high swing frequency), zones appeared and disappeared rapidly, leaving at most one visible.

Fix: Array-Based Multi-Zone Tracking

Converted from scalar to array-based tracking (lines 2056-2137):

Component	Before (scalar)	After (array)
Zone storage	<code>var float demandTop = na</code>	<code>var array<float> dTop = array.new_float(0)</code>
Detection	<code>demandBot := swLow (overwrites)</code>	<code>array.push(dBot, swLow)</code> (accumulates)
Max zones	1 per type	10 per type (oldest auto-trimmed)
Invalidation	Single if check	Reverse while loop over all zones
Rendering	1 <code>box.new()</code> per type	<code>for</code> loop draws all active zones

What Changed (Line-by-Line)

Lines	Change
2035-2054	Updated section header + documentation
2056	Added <code>int maxZones = 10</code> cap
2058-2068	Replaced 10 scalar <code>var</code> declarations with 8 <code>var array<></code> declarations
2072-2083	Demand detection: <code>array.push()</code> + <code>array.shift()</code> trim instead of scalar overwrite
2086-2097	Supply detection: same array-push pattern
2099-2107	Demand invalidation: reverse-iteration while loop with <code>array.remove()</code>
2109-2117	Supply invalidation: same reverse-iteration pattern
2119-2137	Rendering: <code>for</code> loops drawing ALL active zone boxes (was 1 each)

What Was NOT Changed

Component	Status
Zone detection trigger	✓ Same: <code>swLow/swHigh + displacement > 0.5</code>
Zone height calculation	✓ Same: <code>atrValue * 0.5</code>
Zone invalidation threshold	✓ Same: <code>atrValue * 0.3</code> beyond boundary
Box styling (colors, borders, text)	✓ Identical <code>box.new()</code> params
H4 Premium/Discount background zones	✓ Completely separate system, untouched
<code>alert()</code> architecture	✓ Zero changes
<code>alertcondition()</code> messages	✓ Zero changes
Signal firing logic	✓ Zero changes
SL/TP calculations	✓ Zero changes
HUD table	✓ Zero changes
All other files	✓ Zero changes

Expected Behavior After Fix

On **5M charts**:

- ✓ Multiple filled **Supply** zones (orange, "● SUPPLY" label)
- ✓ Multiple filled **Demand** zones (green, "● DEMAND" label)
- ✓ Up to 10 active zones per type simultaneously
- ✓ Zones auto-invalidate when price breaks through
- ✓ Oldest zones trimmed when >10 accumulate

On **H4 charts**:

- ✓ H4 Premium/Discount background zones unchanged
- ✓ Supply/Demand zones also render (fewer due to slower swing detection)

Deployment Verification

```
{"service": "SMC Performance Tracker", "status": "ok", "version": "v17.54"}
```

Next Step for User

Paste the updated Pine Script into TradingView to see the multi-zone rendering. The `.pine` file in the repo is the source of truth — copy it into TradingView's Pine Editor and apply to a 5M chart.